GUI Front End for RASOrbit - 1

5-13-2026

To get started with RASOrbit Input, I need the following:

1. A correct input file with all values present (to ensure read/write fidelity)
2. For each input:
    1. UI label
    2. internal name / short ID
    3. required/optional
    4. default value (if any)
    5. min value (if any)
    6. max value (if any)
    7. step value if discrete (integer or fixed fraction digits)
    8. type of input if not number (check box, radio buttons, combo box)
    9. dependencies/constraints on other fields (if any)
3. If applicable, groups into which the inputs should be organized

Optionally, if you have some sloppy or mistaken input files that people have created in Notepad, I can ensure they are read and written in the correct format. For this, I would need a set of pairs of sloppy and fixed files so I can ensure everything is interpreted correctly. Note that I can't run RASOrbit, so I need to test by ensuring that written files match known good ones.

From your May 10 e-mail.

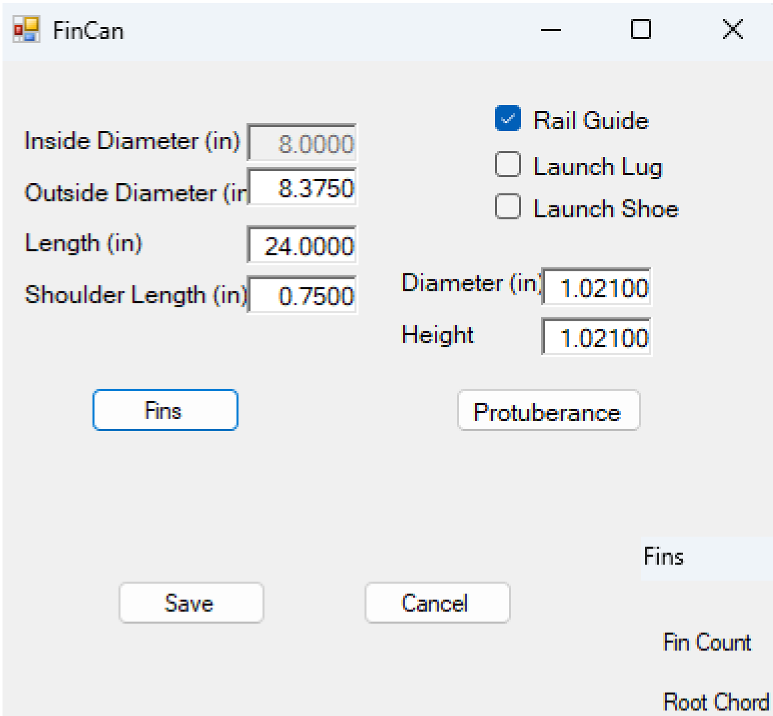
Hi John,

I put this at the front of the slides, so it wouldn't get buried at the back of the slides.
I think in the slides I have addressed most of these, except for:

Optionally, if you have some sloppy or mistaken input files that people have created in Notepad, I can ensure they are read and written in the correct format. For this, I would need a set of pairs of sloppy and fixed files so I can ensure everything is interpreted correctly. Note that I can't run RASOrbit, so I need to test by ensuring that written files match known good ones.

I'll include some example messed-up Input Files.

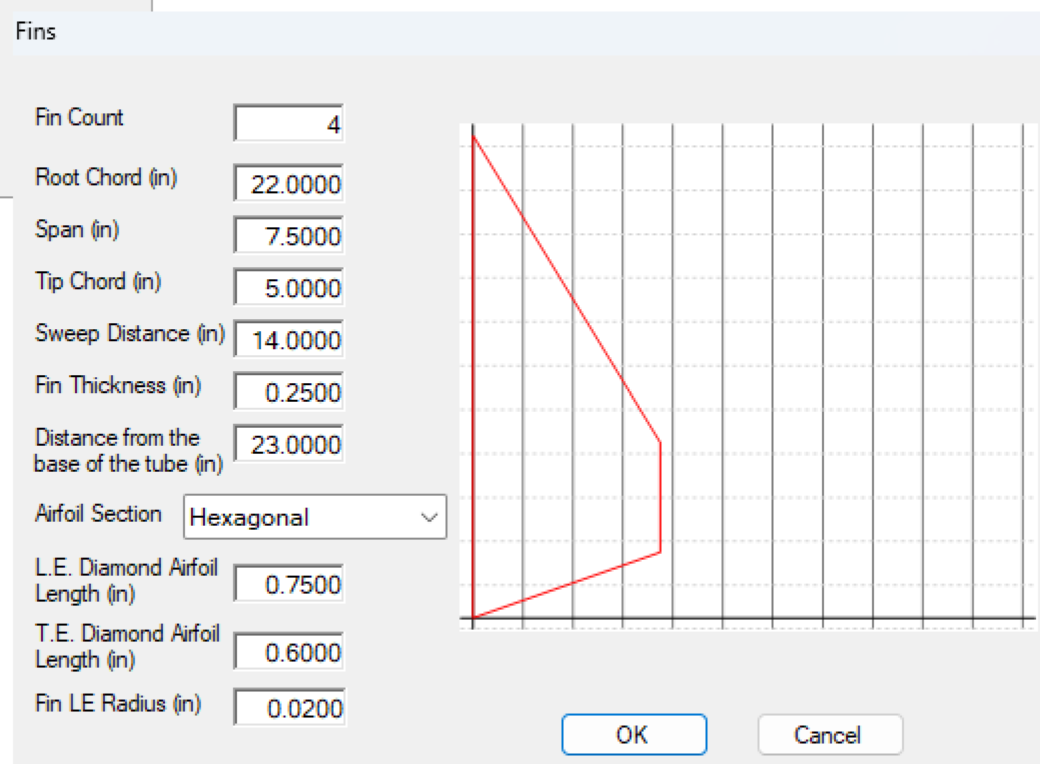
Although at some point, if they have messed it up enough, then we should leave the Input Blank.



For the Graphical User Interface (GUI) I'd like a similar look and feel as RASAero II.

Run RASAero II, and take a look at the GUI Input Windows, that's the kind of look and feel I'd like.

I'd like the same font as used in RASAero II, but <u>I'd like the font to be made bigger</u>. See the slides which start on Slide 6.



```
MINOTAUR I SMALL FAIRING 450KM CIRCULAR POLAR - PAYLOAD TO ORBIT = 897.0 LBS
 4         1         1
 5
-15.0     -5.0      0.0       5.0       15.0
15
0.0       0.1       0.25      0.5       0.9       0.95      1.05      1.25
1.50      2.0       3.0       4.0       5.0       7.5       10.0
-0.932    -0.223    0.0       0.223     0.932
-0.94     -0.223    0.0       0.223     0.94
-0.954    -0.224    0.0       0.224     0.954
-0.977    -0.225    0.0       0.225     0.977
-1.032    -0.226    0.0       0.226     1.032
-1.072    -0.229    0.0       0.229     1.072
-1.243    -0.243    0.0       0.243     1.243
-1.620    -0.342    0.0       0.342     1.620
-1.770    -0.369    0.0       0.369     1.770
-2.001    -0.378    0.0       0.378     2.001
-2.05     -0.371    0.0       0.371     2.05
-1.843    -0.357    0.0       0.357     1.843
-1.75     -0.364    0.0       0.364     1.75
-1.694    -0.385    0.0       0.385     1.694
-1.687    -0.377    0.0       0.377     1.687
0.243     0.243     0.243     0.243     0.243
0.199     0.199     0.199     0.199     0.199
0.187     0.187     0.187     0.187     0.187
0.179     0.179     0.179     0.179     0.179
0.197     0.197     0.197     0.197     0.197
0.224     0.224     0.224     0.224     0.224
0.273     0.273     0.273     0.273     0.273
0.748     0.748     0.748     0.748     0.748
0.692     0.692     0.692     0.692     0.692
0.627     0.627     0.627     0.627     0.627
0.549     0.549     0.549     0.549     0.549
0.496     0.496     0.496     0.496     0.496
0.467     0.464     0.463     0.464     0.467
0.443     0.429     0.427     0.429     0.443
0.446     0.428     0.426     0.428     0.446
```

The GUI reads, modifies, and writes the RASOrbit Input File, which is the .dat file.

That's all it does. It reads an existing .dat file, or modifies an existing .dat file, or creates a new .dat file.

Two example .dat Input Files have been included in the zip file:

Data-10A – US Units.dat
Data-10B – Metric Units.dat

For Now we're only working on the first one-half of the RASOrbit Input File.

We'll have to agree on an approach, we'll probably be iterating and changing a few things, maybe adjusting the overall look and feel, so I thought it made sense to work on only the first one-half of the Input File first.

Once the first one-half of the Input File is in good shape, then we'll move on to the rest of the Input File.

The slides which follow cover only the first one-half of the Input File.

The GUI can Test-Read just the first one-half of the two example Input Files, and can Test-Write just the first one-half of an example Input File.

1st Input Page

Title

Allow Up To 80 Characters.
Do Not Allow More Than 80 Characters.

Line 1

```
MINOTAUR I SMALL FAIRING 450KM CIRCULAR POLAR - PAYLOAD TO ORBIT = 897.0 LBS
```

Column 1

Can Go Out to Column 80

1st Input Page (Continued)

Title

Number of Stages (1-4) — **First number, must be 1-4, is an Integer**

Units

SI (Metric) Units — **If checked second number is 0 (Integer)** — **SI (Metric) Units**

English Units — **If checked second number is 1 (Integer)** — **English Units**

Aerodynamic Data

CL, CD, and Center of Pressure (CP) — **If checked third number is 0 (Integer)** — **CL, CD, CP**

CN, CA, and CP — **If checked third number is 1 (Integer)** — **CN, CA, CP**

Save | Cancel

```
MINOTAUR I SMALL FAIRING 450KM CIRCULAR POLAR - PAYLOAD TO ORBIT = 897.0 LBS
 4         1         1
```

Line 2

First Number
Column 2
Integer

Second Number
Column 12
Integer

Third Number
Column 22
Integer

At this point there can be four branches:

SI (Metric) Units
CL, CD, CP

English Units
CL, CD, CP

SI (Metric) Units
CN, CA, CP

English Units
CN, CA, CP

The Numbers get Saved in the .dat Input File the same. RASOrbit knows the Units and whether it is CL, CD, CP or CN, CA, CP by the Inputs on Line 2, but the Numbers get Saved in the .dat Input File the same.

The Number also get Read from the .dat Input File the same, whether it is:
1) SI (Metric) Units or English Units
2) CL, CD, CP, or CN, CA, CP

RASOrbit determines this after Reading Line 2.

The Changes are What is Displayed in the GUI, which is detailed in the Subsequent Slides.

English Units selected

CN, CA, CP selected

2nd Input Page - English Units – CN, CA, CP

Aerodynamic Data

Make this Bold

Notes on Aerodynamic Data:

For Liquid Rocket Engine or Solid Rocket Motor Powered Stages Aerodynamic Data is Power-On Aerodynamic Data. Power-Off Delta Axial Force Increment is then Added for Coasting Phase of Flight.

Power-Off Delta Axial Force Increment is Assumed Constant with Angle of Attack.

If Stage is Unpowered, then Aerodynamic Data is Power-Off Aerodynamic Data. Delta Axial Force Increment for Power-Off is Set to Zero.

OK

Cancel

3rd Input Page – English Units – CN, CA, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make these Bold

Aerodynamic Data

Number of Angles of Attack (Min 3, Max 8) 8

Minimum of 3, Maximum of 8, is an Integer

Angles of Attack

0.0000	0.0000	0.0000	0.0000	0.0000

Will dynamically update based on number of Angles of Attack. Numbers can be Negative. Allow up to 99.0000, -99.0000 Probably one row.

Number of Mach Numbers (Max 15) 15

Minimum of 2, Maximum of 15, Integer

Mach Numbers

0.0000	0.0000	0.0000	0.0000	0.0000

Will dynamically update based on number of Mach Numbers. May need two rows, put 8 on first row. Allow up to 99.0000

Normal Force Coefficient (CN)

Make this Bold

Angles of Attack

Will dynamically update based on number of Angles of Attack and number of Mach Numbers. Numbers can be Negative. Allow up to 99.0000, -99.0000

Mach Number	-2.5	0.0	2.5	5.0	7.5	10.0	15.0
0.25	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.75	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Save

Cancel

Number of Angles of Attack, Max is 8.
_8 Must start in Column 2

Number of Mach Numbers.
If less than 10, _8, _9, etc., start in Column 2.
If greater than 10, then 10, 11, 15,
Start in Column 1.

```
MINOTAUR I SMALL FAIRING 450KM CIRCULAR POLAR - PAYLOAD TO ORBIT = 897.0 LBS
 4         1         1
 5
-15.0     -5.0      0.0       5.0       15.0
15
0.0       0.1       0.25      0.5       0.9       0.95      1.05      1.25
1.50      2.0       3.0       4.0       5.0       7.5       10.0
-0.932    -0.223    0.0       0.223     0.932
-0.94     -0.223    0.0       0.223     0.94
-0.954    -0.224    0.0       0.224     0.954
-0.977    -0.225    0.0       0.225     0.977
-1.032    -0.226    0.0       0.226     1.032
-1.072    -0.229    0.0       0.229     1.072
-1.243    -0.243    0.0       0.243     1.243
-1.620    -0.342    0.0       0.342     1.620
-1.770    -0.369    0.0       0.369     1.770
-2.001    -0.378    0.0       0.378     2.001
-2.05     -0.371    0.0       0.371     2.05
-1.843    -0.357    0.0       0.357     1.843
-1.75     -0.364    0.0       0.364     1.75
-1.694    -0.385    0.0       0.385     1.694
-1.687    -0.377    0.0       0.377     1.687
```

Angles of Attack

Mach Numbers

Mach
Number
0.0
0.1
0.25
0.5
Etc.

CN

Negative Sign or First Number
Must Be In These Columns

Columns 1 11 21 31 41 51 61 71

Angles of Attack
-15.0 -5.0 0.0 5.0 15.0

4th Input Page - English Units – CN, CA, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make these Bold

Aerodynamic Data

Axial Force Coefficient (CA)

Make this Bold

Will dynamically update based on number of Angles of Attack and number of Mach Numbers.
Allow for numbers to be Negative (they almost certainly won't be, but allow it).
Allow up to 99.0000, -99.0000

Angles of Attack

Mach Number	-2.5	0.0	2.5	5.0	7.5	10.0	15.0
0.25	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.75	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Save

Cancel

Mach Number
0.0
0.1
0.25
0.5
Etc.

CN

-0.932	-0.223	0.0	0.223	0.932
-0.94	-0.223	0.0	0.223	0.94
-0.954	-0.224	0.0	0.224	0.954
-0.977	-0.225	0.0	0.225	0.977
-1.032	-0.226	0.0	0.226	1.032
-1.072	-0.229	0.0	0.229	1.072
-1.243	-0.243	0.0	0.243	1.243
-1.620	-0.342	0.0	0.342	1.620
-1.770	-0.369	0.0	0.369	1.770
-2.001	-0.378	0.0	0.378	2.001
-2.05	-0.371	0.0	0.371	2.05
-1.843	-0.357	0.0	0.357	1.843
-1.75	-0.364	0.0	0.364	1.75
-1.694	-0.385	0.0	0.385	1.694
-1.687	-0.377	0.0	0.377	1.687

CA

0.243	0.243	0.243	0.243	0.243
0.199	0.199	0.199	0.199	0.199
0.187	0.187	0.187	0.187	0.187
0.179	0.179	0.179	0.179	0.179
0.197	0.197	0.197	0.197	0.197
0.224	0.224	0.224	0.224	0.224
0.273	0.273	0.273	0.273	0.273
0.748	0.748	0.748	0.748	0.748
0.692	0.692	0.692	0.692	0.692
0.627	0.627	0.627	0.627	0.627
0.549	0.549	0.549	0.549	0.549
0.496	0.496	0.496	0.496	0.496
0.467	0.464	0.463	0.464	0.467
0.443	0.429	0.427	0.429	0.443
0.446	0.428	0.426	0.428	0.446

Columns 1 11 21 31 41 51 61 71

Angles of Attack
-15.0 -5.0 0.0 5.0 15.0

5th Input Page - English Units – CN, CA, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make these Bold

Aerodynamic Data

Center of Pressure – CP – (inches from nose tip)

Make this Bold

Angles of Attack

Will dynamically update based on number of Angles of Attack and number of Mach Numbers.
Allow for numbers to be Negative (they almost certainly won't be, but allow it).
Allow up to 9999.0000, -9999.0000

Mach Number	-2.5	0.0	2.5	5.0	7.5	10.0	15.0
0.25	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000
0.5	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000
0.75	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000

Save

Cancel

Mach Number
0.0
0.1
0.25
0.5
Etc.

CA

0.243	0.243	0.243	0.243	0.243
0.199	0.199	0.199	0.199	0.199
0.187	0.187	0.187	0.187	0.187
0.179	0.179	0.179	0.179	0.179
0.197	0.197	0.197	0.197	0.197
0.224	0.224	0.224	0.224	0.224
0.273	0.273	0.273	0.273	0.273
0.748	0.748	0.748	0.748	0.748
0.692	0.692	0.692	0.692	0.692
0.627	0.627	0.627	0.627	0.627
0.549	0.549	0.549	0.549	0.549
0.496	0.496	0.496	0.496	0.496
0.467	0.464	0.463	0.464	0.467
0.443	0.429	0.427	0.429	0.443
0.446	0.428	0.426	0.428	0.446
460.0	460.0	460.0	460.0	460.0
460.0	460.0	460.0	460.0	460.0
460.0	460.0	460.0	460.0	460.0
460.0	460.0	460.0	460.0	460.0
460.0	460.0	460.0	460.0	460.0
470.0	470.0	470.0	470.0	470.0
480.0	480.0	480.0	480.0	480.0
475.0	475.0	475.0	475.0	475.0
470.0	470.0	470.0	470.0	470.0
465.0	465.0	465.0	465.0	465.0
460.0	460.0	460.0	460.0	460.0
455.0	455.0	455.0	455.0	455.0
450.0	450.0	450.0	450.0	450.0
445.0	445.0	445.0	445.0	445.0
440.0	440.0	440.0	440.0	440.0

CP

Columns 1 11 21 31 41 51 61 71

Angles of Attack
-15.0 -5.0 0.0 5.0 15.0

6th Input Page - English Units – CN, CA, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make these Bold

Aerodynamic Data

See Here First.

Power-Off Delta Axial Force Coefficient

If 8 Mach Numbers or less, use one row.
If more than 8 Mach Numbers, use two rows, with 8 Mach Numbers in the First Row.

Mach Number

0.0000	0.1000	0.2500	0.5000	0.9000	0.9500	1.0500	1.25000
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Mach Number

1.5000	2.0000	3.0000	4.0000	5.0000	7.5000	10.0000
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Positive Power-Off Delta Axial Force Coefficient Means Power-Off Axial Force Coefficient Higher than Power-On Axial Force Coefficient.

Power-Off Delta Axial Force Increment is Assumed Constant with Angle of Attack.

The list of Mach Numbers is Auto-Populated from the original set of Mach Numbers used for the CN, CA, and CP Aero Data.

The list of Mach Numbers and the number of Input Boxes will have to dynamically update based on the original list of Mach Numbers from the original input.

Values can be Positive or Negative. Allow up to 99.0000, -99.0000

Save Cancel

460.0	460.0	460.0	460.0	460.0			
460.0	460.0	460.0	460.0	460.0			
460.0	460.0	460.0	460.0	460.0			
460.0	460.0	460.0	460.0	460.0			
460.0	460.0	460.0	460.0	460.0			
470.0	470.0	470.0	470.0	470.0			
480.0	480.0	480.0	480.0	480.0			
475.0	475.0	475.0	475.0	475.0			
470.0	470.0	470.0	470.0	470.0			
465.0	465.0	465.0	465.0	465.0			
460.0	460.0	460.0	460.0	460.0			
455.0	455.0	455.0	455.0	455.0			
450.0	450.0	450.0	450.0	450.0			
445.0	445.0	445.0	445.0	445.0			
440.0	440.0	440.0	440.0	440.0			
0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
0.05	0.05	0.05	0.05	0.05	0.05	0.05	

CP

Power-Off Delta Axial Force Coefficient Data

Columns 1 11 21 31 41 51 61 71

Mach Numbers

0.0	0.1	0.25	0.5	0.9	0.95	1.05	1.25
1.50	2.0	3.0	4.0	5.0	7.5	10.0	

7th Input Page - English Units – CN, CA, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make this Bold

Number of Weight Points for Center of Gravity (CG) and Inertia Data 10 Minimum of 2, Maximum of 10, is an Integer

Will dynamically update based on number of Weight Breakpoints.
Up to the Limit (10), make one row.

Weight (lbs)

0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Center of Gravity (CG) - (inches from nose tip)

0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Pitch Inertia (slug-ft^2)

0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

If do not wish to calculate Thrust Vector Angle, then Enter Zeros for CG.
Flight simulation will still run. Thrust Vector Angle will be Zero.

If do not wish to calculate Time-to-Double-Amplitude, then Enter Zeros for CG and Pitch Inertia.
Flight simulation will still run. Time-to-Double-Amplitude will not be calculated.

Save Cancel

```
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
470.0     470.0     470.0     470.0     470.0
480.0     480.0     480.0     480.0     480.0
475.0     475.0     475.0     475.0     475.0
470.0     470.0     470.0     470.0     470.0
465.0     465.0     465.0     465.0     465.0
460.0     460.0     460.0     460.0     460.0
455.0     455.0     455.0     455.0     455.0
450.0     450.0     450.0     450.0     450.0
445.0     445.0     445.0     445.0     445.0
440.0     440.0     440.0     440.0     440.0
0.05      0.05      0.05      0.05      0.05      0.05      0.05      0.05
0.05      0.05      0.05      0.05      0.05      0.05      0.05
 2
0.0       0.0
0.0       0.0
0.0       0.0
0.0       0.0       0.0
```

CP

Number of Weight Breakpoints.

Power-Off Delta Axial Force Coefficient Data

Weight Breakpoints.

Center of Gravity (CG) Data (inches)

Pitch Inertia Data (slugs-ft^2)

Columns 1 11 21 31 41 51 61 71

Mach Numbers

0.0	0.1	0.25	0.5	0.9	0.95	1.05	1.25
1.50	2.0	3.0	4.0	5.0	7.5	10.0	

8th Input Page - English Units – CN, CA, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make this Bold

Thrust Vectoring Gimbal Location – (inches from nose tip)

0.0000

Percent of Thrust for Pitch Thrust Vectoring – Enter 1.0 for 100%, 0.5 for 50%

0.0000

Maximum Thrust Vector Angle (deg)

0.0000

If Stage is Unpowered, Enter Zero for Thrust Vectoring Gimbal Location, Percent of Thrust for Pitch Thrust Vectoring, and Maximum Thrust Vector Angle.

If do not wish to calculate Thrust Vector Angle, then Enter Zero for Thrust Vectoring Gimbal Location, Percent of Thrust for Pitch Thrust Vectoring, and Maximum Thrust Vector Angle.

Save

Cancel

```
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
470.0     470.0     470.0     470.0     470.0
480.0     480.0     480.0     480.0     480.0
475.0     475.0     475.0     475.0     475.0
470.0     470.0     470.0     470.0     470.0
465.0     465.0     465.0     465.0     465.0
460.0     460.0     460.0     460.0     460.0
455.0     455.0     455.0     455.0     455.0
450.0     450.0     450.0     450.0     450.0
445.0     445.0     445.0     445.0     445.0
440.0     440.0     440.0     440.0     440.0
0.05      0.05      0.05      0.05      0.05      0.05      0.05      0.05
0.05      0.05      0.05      0.05      0.05      0.05      0.05
 2
0.0       0.0
0.0       0.0
0.0       0.0
0.0       0.0       0.0
```

CP

Power-Off Delta Axial Force Coefficient Data

Thrust Vectoring Gimbal Location (inches from nose tip)

Percent of Thrust for Pitch Thrust Vectoring

Maximum Thrust Vector Angle (deg)

Columns 1 11 21 31 41 51 61 71

```
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
460.0     460.0     460.0     460.0     460.0
470.0     470.0     470.0     470.0     470.0
480.0     480.0     480.0     480.0     480.0
475.0     475.0     475.0     475.0     475.0
470.0     470.0     470.0     470.0     470.0
465.0     465.0     465.0     465.0     465.0
460.0     460.0     460.0     460.0     460.0
455.0     455.0     455.0     455.0     455.0
450.0     450.0     450.0     450.0     450.0
445.0     445.0     445.0     445.0     445.0
440.0     440.0     440.0     440.0     440.0
0.05      0.05      0.05      0.05      0.05      0.05      0.05      0.05
0.05      0.05      0.05      0.05      0.05      0.05      0.05
 2
0.0       0.0
0.0       0.0
0.0       0.0
0.0       0.0       0.0
```

End of Stage-1 Input Data

Beginning of Stage-2 Input Data

```
 5
-15.0     -5.0      0.0       5.0       15.0
```

Angles of Attack

```
 7
3.0       4.0       5.0       7.5       10.0      15.0      20.0
```

Mach Numbers

Mach Number
0.0
0.1
0.25
0.5
Etc.

```
-1.232    -0.222    0.0       0.222     1.232
-1.148    -0.228    0.0       0.228     1.148
-1.109    -0.238    0.0       0.238     1.109
-1.094    -0.257    0.0       0.257     1.094
-1.102    -0.257    0.0       0.257     1.102
-1.121    -0.253    0.0       0.253     1.121
-1.136    -0.255    0.0       0.255     1.136
```

CN

Inputs are Repeated for Next Stage

SI (Metric) Units selected

CN, CA, CP selected

(Changes in Blue)

5th Input Page – SI (Metric) Units – CN, CA, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make these Bold

Aerodynamic Data

Center of Pressure – CP – (meters from nose tip)

Make this Bold

Will dynamically update based on number of Angles of Attack and number of Mach Numbers.
Allow for numbers to be Negative (they almost certainly won't be, but allow it).
Allow up to 9999.0000, -9999.0000

Angles of Attack

Mach Number	-2.5	0.0	2.5	5.0	7.5	10.0	15.0
0.25	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000
0.5	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000
0.75	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000

Save Cancel

0.243	0.243	0.243	0.243	0.243
0.199	0.199	0.199	0.199	0.199
0.187	0.187	0.187	0.187	0.187
0.179	0.179	0.179	0.179	0.179
0.197	0.197	0.197	0.197	0.197
0.224	0.224	0.224	0.224	0.224
0.273	0.273	0.273	0.273	0.273
0.748	0.748	0.748 CA	0.748	0.748
0.692	0.692	0.692	0.692	0.692
0.627	0.627	0.627	0.627	0.627
0.549	0.549	0.549	0.549	0.549
0.496	0.496	0.496	0.496	0.496
0.467	0.464	0.463	0.464	0.467
0.443	0.429	0.427	0.429	0.443
0.446	0.428	0.426	0.428	0.446
11.684	11.684	11.684	11.684	11.684
11.684	11.684	11.684	11.684	11.684
11.684	11.684	11.684	11.684	11.684
11.684	11.684	11.684	11.684	11.684
11.684	11.684	11.684	11.684	11.684
11.938	11.938	11.938	11.938	11.938
12.192	12.192	12.192 CP	12.192	12.192
12.065	12.065	12.065	12.065	12.065
11.938	11.938	11.938	11.938	11.938
11.811	11.811	11.811	11.811	11.811
11.684	11.684	11.684	11.684	11.684
11.557	11.557	11.557	11.557	11.557
11.43	11.43	11.43	11.43	11.43
11.303	11.303	11.303	11.303	11.303
11.176	11.176	11.176	11.176	11.176

Mach Number
0.0
0.1
0.25
0.5
Etc.

Note that the Numbers Change, as the Units are Now SI (Metric) rather than English Units. CP is Now in Meters, rather than in Inches. But the Numbers are Stored the Same.

Relative to the RASOrbit Input File Loaded by RASOrbit Loader, You Just Read the Inputs and Fill In the Inputs. RASOrbit Knows from <u>Line 2</u> of the Input File whether the Inputs are SI (Metric) or English Units.

Columns 1 11 21 31 41 51 61 71

Angles of Attack
-15.0 -5.0 0.0 5.0 15.0

7th Input Page – SI (Metric) Units – CN, CA, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make this Bold

Number of Weight Points for Center of Gravity (CG) and Inertia Data [10] Minimum of 2, Maximum of 10, is an Integer

Will dynamically update based on number of Weight Breakpoints.
Up to the Limit (10), make one row.

Weight (kg)

0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Center of Gravity (CG) - (meters from nose tip)

0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Pitch Inertia (kg-m^2)

0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

If do not wish to calculate Thrust Vector Angle, then Enter Zeros for CG.
Flight simulation will still run. Thrust Vector Angle will be Zero.

If do not wish to calculate Time-to-Double-Amplitude, then Enter Zeros for CG and Pitch Inertia.
Flight simulation will still run. Time-to-Double-Amplitude will not be calculated.

Save Cancel

```
11.684    11.684    11.684    11.684    11.684
11.684    11.684    11.684    11.684    11.684
11.684    11.684    11.684    11.684    11.684
11.684    11.684    11.684    11.684    11.684
11.684    11.684    11.684    11.684    11.684
11.938    11.938    11.938    11.938    11.938
12.192    12.192    12.192    12.192    12.192
12.065    12.065    12.065    12.065    12.065
11.938    11.938    11.938    11.938    11.938
11.811    11.811    11.811    11.811    11.811
11.684    11.684    11.684    11.684    11.684
11.557    11.557    11.557    11.557    11.557
11.43     11.43     11.43     11.43     11.43
11.303    11.303    11.303    11.303    11.303
11.176    11.176    11.176    11.176    11.176
0.05      0.05      0.05      0.05      0.05      0.05      0.05      0.05
0.05      0.05      0.05      0.05      0.05      0.05      0.05
 2
0.0       0.0
0.0       0.0
0.0       0.0
0.0       0.0       0.0
```

CP

Note that the Numbers Change, as the Units are Now SI (Metric) rather than English Units. But the Numbers Are Stored the Same.

Relative to the RASOrbit Input File Loaded by RASOrbit Loader, You Just Read the Inputs and Fill In the Inputs. RASOrbit Knows from Line 2 of the Input File whether the Inputs are SI (Metric) or English Units.

Weight Breakpoints.

Center of Gravity (CG) Data (meters)

Pitch Inertia Data (kg-m^2)

Columns 1 11 21 31 41 51 61 71

Mach Numbers

0.0	0.1	0.25	0.5	0.9	0.95	1.05	1.25
1.50	2.0	3.0	4.0	5.0	7.5	10.0	

8th Input Page – SI (Metric) Units – CN, CA, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make this Bold

Thrust Vectoring Gimbal Location – (meters from nose tip)

0.0000

Percent of Thrust for Pitch Thrust Vectoring – Enter 1.0 for 100%, 0.5 for 50%

0.0000

Maximum Thrust Vector Angle (deg)

0.0000

If Stage is Unpowered, Enter Zero for Thrust Vectoring Gimbal Location, Percent of Thrust for Pitch Thrust Vectoring, and Maximum Thrust Vector Angle.

If do not wish to calculate Thrust Vector Angle, then Enter Zero for Thrust Vectoring Gimbal Location, Percent of Thrust for Pitch Thrust Vectoring, and Maximum Thrust Vector Angle.

Save

Cancel

English Units selected

CL, CD, CP selected

(Changes in Blue)

2nd Input Page - English Units – CL, CD, CP

Aerodynamic Data ← Make this Bold

Notes on Aerodynamic Data:

For Liquid Rocket Engine or Solid Rocket Motor Powered Stages Aerodynamic Data is Power-On Aerodynamic Data. Power-Off Delta Drag Coefficient Increment is then Added for Coasting Phase of Flight.

~~Power-Off Delta Axial Force Increment is Assumed Constant with Angle of Attack.~~

If Stage is Unpowered, then Aerodynamic Data is Power-Off Aerodynamic Data. Delta Drag Coefficient Increment for Power-Off is Set to Zero.

OK   Cancel

3rd Input Page – English Units – CL, CD, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make these Bold

Aerodynamic Data

Number of Angles of Attack (Min 3, Max 8) 8

Minimum of 3, Maximum of 8, is an Integer

Angles of Attack

0.0000	0.0000	0.0000	0.0000	0.0000

Will dynamically update based on number of Angles of Attack. Numbers can be Negative. Allow up to 99.0000, -99.0000 Probably one row.

Number of Mach Numbers (Max 15) 15

Minimum of 2, Maximum of 15, Integer

Mach Numbers

0.0000	0.0000	0.0000	0.0000	0.0000

Will dynamically update based on number of Mach Numbers. May need two rows, put 8 on first row. Allow up to 99.0000

Lift Coefficient (CL)

Make this Bold

Angles of Attack

Will dynamically update based on number of Angles of Attack and number of Mach Numbers. Numbers can be Negative. Allow up to 99.0000, -99.0000

Mach Number	-2.5	0.0	2.5	5.0	7.5	10.0	15.0
0.25	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.75	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Save Cancel

Number of Angles of Attack, Max is 8.
_8 Must start in Column 2

Number of Mach Numbers.
If less than 10, _8, _9, etc., start in Column 2.
If greater than 10, then 10, 11, 15,
Start in Column 1.

```
MINOTAUR I SMALL FAIRING 450KM CIRCULAR POLAR - PAYLOAD TO ORBIT = 897.0 LBS
 4        1         1
 5
-15.0     -5.0      0.0       5.0       15.0
15
0.0       0.1       0.25      0.5       0.9       0.95      1.05      1.25
1.50      2.0       3.0       4.0       5.0       7.5       10.0
-0.932    -0.223    0.0       0.223     0.932
-0.94     -0.223    0.0       0.223     0.94
-0.954    -0.224    0.0       0.224     0.954
-0.977    -0.225    0.0       0.225     0.977
-1.032    -0.226    0.0       0.226     1.032
-1.072    -0.229    0.0       0.229     1.072
-1.243    -0.243    0.0       0.243     1.243
-1.620    -0.342    0.0       0.342     1.620
-1.770    -0.369    0.0       0.369     1.770
-2.001    -0.378    0.0       0.378     2.001
-2.05     -0.371    0.0       0.371     2.05
-1.843    -0.357    0.0       0.357     1.843
-1.75     -0.364    0.0       0.364     1.75
-1.694    -0.385    0.0       0.385     1.694
-1.687    -0.377    0.0       0.377     1.687
```

Angles of Attack

Mach Numbers

Mach Number
0.0
0.1
0.25
0.5
Etc.

CL

Negative Sign or First Number
Must Be In These Columns

Columns 1 11 21 31 41 51 61 71

Angles of Attack
-15.0 -5.0 0.0 5.0 15.0

4th Input Page - English Units – CL, CD, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make these Bold

Aerodynamic Data

Drag Coefficient (CD)

Make this Bold

Will dynamically update based on number of Angles of Attack and number of Mach Numbers.
Allow for numbers to be Negative (they almost certainly won't be, but allow it).
Allow up to 99.0000, -99.0000

Angles of Attack

Mach Number	-2.5	0.0	2.5	5.0	7.5	10.0	15.0
0.25	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.75	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Save

Cancel

CL

-0.932	-0.223	0.0	0.223	0.932
-0.94	-0.223	0.0	0.223	0.94
-0.954	-0.224	0.0	0.224	0.954
-0.977	-0.225	0.0	0.225	0.977
-1.032	-0.226	0.0	0.226	1.032
-1.072	-0.229	0.0	0.229	1.072
-1.243	-0.243	0.0	0.243	1.243
-1.620	-0.342	0.0	0.342	1.620
-1.770	-0.369	0.0	0.369	1.770
-2.001	-0.378	0.0	0.378	2.001
-2.05	-0.371	0.0	0.371	2.05
-1.843	-0.357	0.0	0.357	1.843
-1.75	-0.364	0.0	0.364	1.75
-1.694	-0.385	0.0	0.385	1.694
-1.687	-0.377	0.0	0.377	1.687
0.243	0.243	0.243	0.243	0.243
0.199	0.199	0.199	0.199	0.199
0.187	0.187	0.187	0.187	0.187
0.179	0.179	0.179	0.179	0.179
0.197	0.197	0.197	0.197	0.197
0.224	0.224	0.224	0.224	0.224
0.273	0.273	0.273	0.273	0.273
0.748	0.748	0.748	0.748	0.748
0.692	0.692	0.692	0.692	0.692
0.627	0.627	0.627	0.627	0.627
0.549	0.549	0.549	0.549	0.549
0.496	0.496	0.496	0.496	0.496
0.467	0.464	0.463	0.464	0.467
0.443	0.429	0.427	0.429	0.443
0.446	0.428	0.426	0.428	0.446

CD

Mach Number
0.0
0.1
0.25
0.5
Etc.

Columns 1 11 21 31 41 51 61 71

Angles of Attack
-15.0 -5.0 0.0 5.0 15.0

5th Input Page - English Units – CL, CD, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make these Bold

Aerodynamic Data

Center of Pressure – CP – (inches from nose tip)

Make this Bold

Angles of Attack

Will dynamically update based on number of Angles of Attack and number of Mach Numbers.
Allow for numbers to be Negative (they almost certainly won't be, but allow it).
Allow up to 9999.0000, -9999.0000

Mach Number	-2.5	0.0	2.5	5.0	7.5	10.0	15.0
0.25	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000
0.5	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000
0.75	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000	0000.0000

No Changes

Save Cancel

0.243	0.243	0.243	0.243	0.243
0.199	0.199	0.199	0.199	0.199
0.187	0.187	0.187	0.187	0.187
0.179	0.179	0.179	0.179	0.179
0.197	0.197	0.197	0.197	0.197
0.224	0.224	0.224	0.224	0.224
0.273	0.273	0.273	0.273	0.273
0.748	0.748	0.748	0.748	0.748
0.692	0.692	0.692	0.692	0.692
0.627	0.627	0.627	0.627	0.627
0.549	0.549	0.549	0.549	0.549
0.496	0.496	0.496	0.496	0.496
0.467	0.464	0.463	0.464	0.467
0.443	0.429	0.427	0.429	0.443
0.446	0.428	0.426	0.428	0.446
460.0	460.0	460.0	460.0	460.0
460.0	460.0	460.0	460.0	460.0
460.0	460.0	460.0	460.0	460.0
460.0	460.0	460.0	460.0	460.0
460.0	460.0	460.0	460.0	460.0
470.0	470.0	470.0	470.0	470.0
480.0	480.0	480.0	480.0	480.0
475.0	475.0	475.0	475.0	475.0
470.0	470.0	470.0	470.0	470.0
465.0	465.0	465.0	465.0	465.0
460.0	460.0	460.0	460.0	460.0
455.0	455.0	455.0	455.0	455.0
450.0	450.0	450.0	450.0	450.0
445.0	445.0	445.0	445.0	445.0
440.0	440.0	440.0	440.0	440.0

CD

CP

Mach Number
0.0
0.1
0.25
0.5
Etc.

No Changes for CP

Columns 1 11 21 31 41 51 61 71

Angles of Attack
-15.0 -5.0 0.0 5.0 15.0

6th Input Page - English Units – CL, CD, CP

For Stage 1:

This will dynamically update. (Stage 1, Stage 2, etc.)

Make these Bold

Aerodynamic Data

See Here First.

Power-Off Delta Drag Coefficient

If 8 Mach Numbers or less, use one row.
If more than 8 Mach Numbers, use two rows, with 8 Mach Numbers in the First Row.

Mach Number

0.0000	0.1000	0.2500	0.5000	0.9000	0.9500	1.0500	1.25000
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Mach Number

1.5000	2.0000	3.0000	4.0000	5.0000	7.5000	10.0000
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Positive Power-Off Delta Drag Coefficient Means Power-Off Drag Coefficient Higher than Power-On Drag Coefficient.

Power-Off Delta Drag Coefficient is at Zero Degrees Angle of Attack, Converted to Power-Off Delta Axial Force Coefficient, Which is Assumed Constant with Angle of Attack.

The list of Mach Numbers is Auto-Populated from the original set of Mach Numbers used for the CN, CA, and CP Aero Data.

The list of Mach Numbers and the number of Input Boxes will have to dynamically update based on the original list of Mach Numbers from the original input.

Values can be Positive or Negative. Allow up to 99.0000, -99.0000

Save Cancel

460.0	460.0	460.0	460.0	460.0			
460.0	460.0	460.0	460.0	460.0			
460.0	460.0	460.0	460.0	460.0			
460.0	460.0	460.0	460.0	460.0			
460.0	460.0	460.0	460.0	460.0			
470.0	470.0	470.0	470.0	470.0			
480.0	480.0	480.0	480.0	480.0			
475.0	475.0	475.0	475.0	475.0			
470.0	470.0	470.0	470.0	470.0			
465.0	465.0	465.0	465.0	465.0			
460.0	460.0	460.0	460.0	460.0			
455.0	455.0	455.0	455.0	455.0			
450.0	450.0	450.0	450.0	450.0			
445.0	445.0	445.0	445.0	445.0			
440.0	440.0	440.0	440.0	440.0			
0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
0.05	0.05	0.05	0.05	0.05	0.05	0.05	

CP

Power-Off Delta Drag Coefficient Data

Columns 1 11 21 31 41 51 61 71

Mach Numbers

0.0 0.1 0.25 0.5 0.9 0.95 1.05 1.25

1.50 2.0 3.0 4.0 5.0 7.5 10.0

```
460.0    460.0    460.0    460.0    460.0
460.0    460.0    460.0    460.0    460.0
460.0    460.0    460.0    460.0    460.0
460.0    460.0    460.0    460.0    460.0
460.0    460.0    460.0    460.0    460.0
470.0    470.0    470.0    470.0    470.0
480.0    480.0    480.0    480.0    480.0
475.0    475.0    475.0    475.0    475.0
470.0    470.0    470.0    470.0    470.0
465.0    465.0    465.0    465.0    465.0
460.0    460.0    460.0    460.0    460.0
455.0    455.0    455.0    455.0    455.0
450.0    450.0    450.0    450.0    450.0
445.0    445.0    445.0    445.0    445.0
440.0    440.0    440.0    440.0    440.0
0.05     0.05     0.05     0.05     0.05     0.05     0.05     0.05
0.05     0.05     0.05     0.05     0.05     0.05     0.05
 2
0.0      0.0
0.0      0.0
0.0      0.0
0.0      0.0      0.0
```

End of Stage-1 Input Data

Beginning of Stage-2 Input Data

```
 5
-15.0    -5.0     0.0      5.0      15.0
 7
3.0      4.0      5.0      7.5      10.0     15.0     20.0
-1.232   -0.222   0.0      0.222    1.232
-1.148   -0.228   0.0      0.228    1.148
-1.109   -0.238   0.0      0.238    1.109
-1.094   -0.257   0.0      0.257    1.094
-1.102   -0.257   0.0      0.257    1.102
-1.121   -0.253   0.0      0.253    1.121
-1.136   -0.255   0.0      0.255    1.136
```

Angles of Attack

Mach Numbers

Mach Number
0.0
0.1
0.25
0.5
Etc.

CD

Inputs are Repeated for Next Stage